# RAMA SRI SURYA

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## PROFESSIONAL SUMMARY

AI & Autonomous Systems Engineer and Computer Science undergraduate with hands-on experience building deliberative multi-agent reasoning architectures, spatial computer vision pipelines, and production-grade LLM applications. 3x Consecutive Hackathon Champion (Semester 1, Semester 2 / Epochon 2.0, Semester 3 / IIIT Kottayam) with demonstrated engineering rigor in Python, PyTorch, FastAPI, and Google Gemini API. Proven ability to architect resilient distributed systems with sub-millisecond Redis state caching and strict schema validation.

## EDUCATION

**Amrita Vishwa Vidyapeetham** — Kollam, Kerala, India — Expected Graduation: 2029
Bachelor of Technology in Computer Science & Engineering

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Probability & Statistics, Database Management Systems, Deep Learning.
- **Honors:** 3x Consecutive Inter-College Hackathon Winner (Sem 1, Sem 2 / Epochon 2.0, Sem 3 / IIIT Kottayam).

## TECHNICAL SKILLS

**Programming Languages:** Python (Proficient), TypeScript, JavaScript, Java, C, SQL
**AI, ML & Deep Learning:** PyTorch, TensorFlow, Keras 3, Scikit-learn, CNNs, Computer Vision, Person Re-Identification (ReID), Spatial Tracking, MediaPipe 3D
**Generative AI & LLMs:** Google Gemini API (@google/genai SDK), Multi-Agent Systems, Deliberative Reasoning Loops, Prompt Chains, Structured Output Schemas
**Backend & Cloud Systems:** FastAPI, Uvicorn, Pydantic (v2), SQLAlchemy (AsyncIO), Redis, Express.js, Supabase, Vercel Serverless, REST APIs
**DevOps, Data & Tooling:** Docker, Linux/Bash, Git, GitHub Actions, Ruff, Pytest, Pandas, NumPy, Vite, Bun

## FEATURED SYSTEMS & ARCHITECTURE (IN-DEPTH)

Top 4 Flagship Systems

**01. Orbit System — Biometric Edge Vision & Zero-Trust Security** [GitHub] [Live Vercel] — Python, FastAPI, OpenCV, MediaPipe, JavaScript

- Engineered a real-time edge surveillance and automatic lock system utilizing 468-point 3D facial landmark mesh tracking at 60 FPS.
- Implemented sub-50ms distance telemetry calculations and biometric anti-spoofing verification with automated lock/unlock state dispatch.
- Embedded a cybernetic diagnostics CLI providing real-time hardware telemetry and edge surveillance logs.

**02. S2 Legal AI (OmniBots) — Autonomous Multi-Agent Reasoning System** [GitHub] [Live Vercel] — Python, TypeScript, Google Gemini API, Multi-Agent Loops

- Won 1st Place at Epochon 2.0 Hackathon for designing an autonomous multi-agent reasoning architecture that triages legal filings and verifies statutory filing deadlines.
- Implemented a deliberative consensus pattern where specialized agents cross-audit findings against domain constraints prior to returning recommendations, eliminating LLM drift.
- Automated extraction of critical court timelines, jurisdictional filing windows, and structured court memo export.

**03. COSMOS Space Explorer — NASA Telemetry & Audio Planetary Portal** [GitHub] [Live Vercel] — React, TypeScript, NASA Open APIs, Web Speech API, Vite

- Architected real-time deep space telemetry exploration portal streaming high-resolution NASA Curiosity rover imagery, atmospheric pressure telemetry, and mission logs.
- Engineered voice-guided planetary tours with procedural dynamic speech synthesis via the Web Speech API with pitch and rate modulation.
- Integrated orbital distance calculations, planetary ephemerides, and Near-Earth asteroid close-approach tracking.

**04. SmartRetail AI — Autonomous Spatial AI & Retail Risk Engine** [GitHub] [Live Vercel] — Python, FastAPI, Redis AsyncIO, Docker, Pytest

- Engineered a modular computer vision and spatial intelligence platform implementing person re-identification (ReID), real-time customer trajectory tracking, and automated hazard/sweethearting detection.
- Architected non-blocking asynchronous REST services using FastAPI, Pydantic v2, and Redis AsyncIO, achieving sub-millisecond state caching across camera handoffs.
- Built a production-grade multi-stage Docker deployment pipeline verified by comprehensive pytest integration and unit test suites.

## ADDITIONAL PRODUCTION DEPLOYMENTS & SYSTEMS (BRIEF)

Ranks 05–07 Directory

| | | | |
|---|---|---|---|
| 05 | **EduCraft AI** — Multimodal GenAI PDF-to-Course Conversion Engine [GitHub] [Live Vercel] | `TypeScript • Gemini • React 19 • Supabase` |
| 06 | **Agentic AI Omni-Bots** — Industrial Robotics & Defect Quarantine Factory [GitHub] [Live Vercel] | `React • Agentic AI • Web Audio` |
| 07 | **Drive Storage Optimizer** — Cloud Storage Intelligence & Media De-duplication [GitHub] [Live Vercel] | `Node • Express • React • Vercel` |

## ACHIEVEMENTS & HACKATHONS

- **Hackathon Winner (3rd Semester — IIIT Kottayam):** Secured 1st place competing against elite inter-college engineering teams for end-to-end intelligent systems engineering under 36h sprint.
- **Hackathon Winner (2nd Semester / Epochon 2.0):** Won 1st place top honors for architecting autonomous multi-agent deliberative reasoning engine (OmniBots).
- **Hackathon Winner (1st Semester):** Secured 1st place in university-wide hackathon during freshman year for modular software architecture.

## CERTIFICATIONS

- **AIML Certification Program** — IIT Patna
- **Computer Vision with Azure Cognitive Services** — Microsoft
- **Data Analytics Job Simulation** — Deloitte (Forage)
- **AWS Cloud Practitioner Essentials** — Amazon Web Services
- **AI & Generative AI For All** — Infosys Springboard